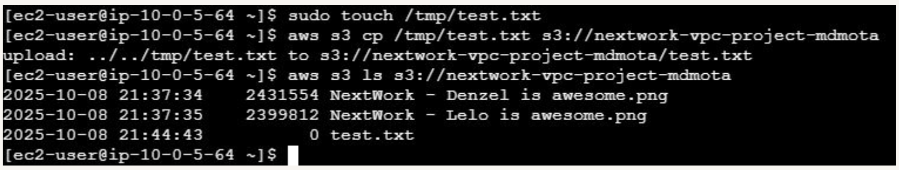
nextwork.org

Access S3 from a VPC

M Mohammed Dawoud Mota



```
[ec2-user@ip-10-0-5-64 ~]$ sudo touch /tmp/test.txt
[ec2-user@ip-10-0-5-64 ~]$ aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-mdmota
upload: ../../tmp/test.txt to s3://nextwork-vpc-project-mdmota/test.txt
[ec2-user@ip-10-0-5-64 ~]$ aws s3 ls s3://nextwork-vpc-project-mdmota
2025-10-08 21:37:34    2431554 NextWork - Denzel is awesome.png
2025-10-08 21:37:35    2399812 NextWork - Lelo is awesome.png
2025-10-08 21:44:43          0 test.txt
[ec2-user@ip-10-0-5-64 ~]$
```

Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual private cloud that lets me launch AWS resources in an isolated network. It is useful because it gives control over networking, security, and access to AWS resources.

How I used Amazon VPC in this project

I used Amazon VPC to create an isolated network where I launched my EC2 instance, allowing it to securely connect and interact with my S3 bucket using access keys and the AWS CLI.

One thing I didn't expect in this project was...

One thing I didn’t expect was that I could upload and manage files in my S3 bucket directly from my EC2 instance using the AWS CLI without needing to use the AWS console.

This project took me...

This project took me 40 minutes to complete.

In the first part of my project...

Step 1 - Architecture set up

In this step, I will create a VPC from scratch and launch an EC2 instance within that VPC.

Step 2 - Connect to my EC2 instance

In this step, I will connect directly to my EC2 instance.

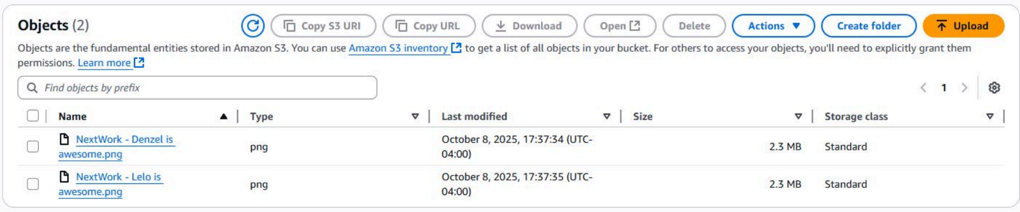
Step 3 - Set up access keys

In this step, I will set up an access key so my EC2 can have access to the AWS environment.

Architecture set up

I started my project by launching a VPC with one public subnet and an EC2 instance inside that subnet.

In this step, I launched an S3 bucket and uploaded two objects into that bucket.

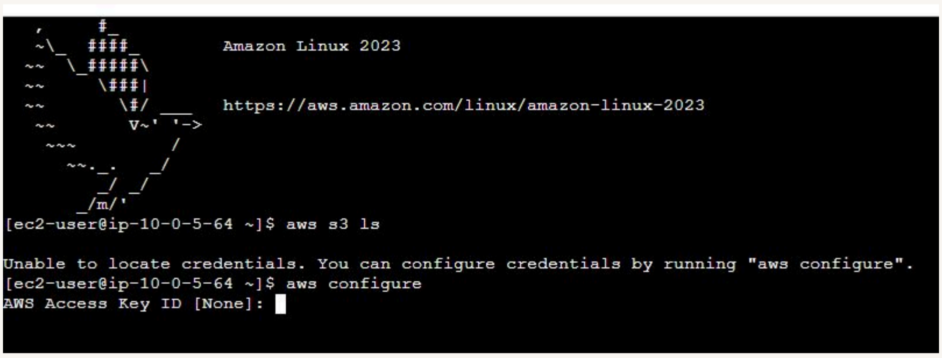


Running CLI commands

AWS CLI is the Command Line Interface tool that lets me manage and interact with AWS services directly from the terminal. It allows me to run commands like listing S3 buckets or uploading files efficiently.

The first command I ran was aws s3 ls which lists all the S3 buckets in my AWS account and helps me verify that my EC2 instance can connect to S3.

The second command I ran was aws configure to set up my EC2 instance with access keys, region, and default output so it could securely interact with AWS services like S3.



Access keys

Credentials

The purpose of the aws configure command was to set up my EC2 instance with access keys, region, and default output format so it could securely interact with AWS services like S3.

Access keys are credentials that consist of an access key ID and a secret access key. They allow my EC2 instance or applications to securely access and manage AWS services in my account.

A secret access key is the password that pairs with the access key ID. It is used to securely authenticate my EC2 instance or applications to access AWS services.

Best practice

The best practice alternative is to create an IAM role with the necessary permissions and attach it to the EC2 instance so it can access AWS services without manually using access keys.

In the second part of my project...

Step 4 - Set up an S3 bucket

In this step, I will launch an S3 bucket. So that my EC2 can have something to access.

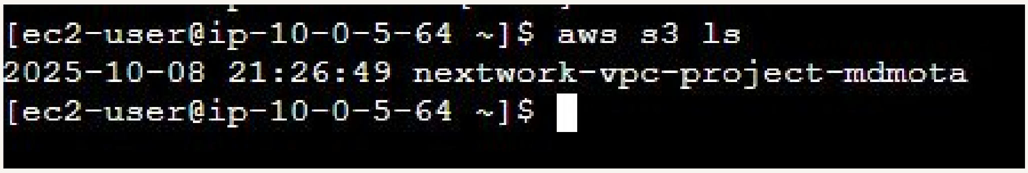
Step 5 - Connecting to my S3 bucket

In this step, I will get my EC2 to interact with my S3 bucket.

Connecting to my S3 bucket

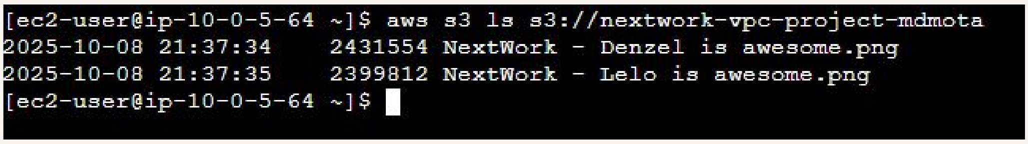
The first command I ran was aws s3 ls which lists all the S3 buckets in my AWS account and helps me verify that my EC2 instance can connect to S3.

My terminal responded with a list of S3 buckets in my account, showing the bucket names and confirming that my EC2 instance was successfully connected and able to access S3.



Connecting to my S3 bucket

The command I ran was aws aws s3 ls s3://nextwork-vpc-project-mdmota to list the objects inside my S3 bucket and verify that the files were accessible from my EC2 instance.



Uploading objects to S3

To upload a new file to my bucket, I first ran the command sudo touch /tmp/test.txt to create a blank .txt file in my EC2 instance.

The second command I ran was aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-mdmota. This command copies the file test.txt from the EC2 instance's tmp folder to my S3 bucket, allowing me to upload files from my instance to S3.

The command aws s3 ls s3://nextwork-vpc-project-mdmota lists all the objects inside my S3 bucket, showing the files I have uploaded and confirming that my EC2 instance can access the bucket.

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